Collisional harvesting of renewed coherence: impedance matching and accessible ergotropy

Noufal Jaseem
Department of Physical Sciences, Indian Institute of Science Education and Research Berhampur, India

Bitan De
Institute of Theoretical Physics, Jagiellonian University in Kraków, Lojasiewicza 11, 30-348 Kraków, Poland

Parvinder Solanki
Department of Physics, Indian Institute of Technology Bombay, Powai, Mumbai 400076, India

Sai Vinjanampathy*
Department of Physics, Indian Institute of Technology Bombay, Powai, Mumbai 400076, India and Centre for Quantum Technologies, National University of Singapore, 3 Science Drive 2, 117543 Singapore, Singapore

Bhaskaran Muralidharan†
Department of Electrical Engineering, Indian Institute of Technology Bombay, Powai, Mumbai 400076, India
(Dated: July 18, 2026)

We study continuous conversion of a renewed quantum coherence into the ergotropy of a tape of qubit batteries. A V-type three-level system carries coherence in a split excited doublet and collides, one by one, with ground-state ancillas through a resonant exchange that conserves bare energy and the intended excitation number. We derive the exact collision map, the stroboscopic fixed point under a renewal bath, and the outgoing qubit ergotropy. The main question is simple: what collision rate maximizes the charging power? We show that the optimum is an impedance match between coherence regrowth and collisional depletion. In the weak-collision limit this gives a broad plateau and a closed law for finite precession of the gap. We also separate total ergotropy from work that can be accessed without a phase reference, and we give the full three-regime result for two energy-dephased copies. The model is a companion theory for coherence harvesting. It is not a claim about restoring electrical transport power.

I. INTRODUCTION

Quantum coherence can change the work that can be drawn from a finite system. Ergotropy and work that can actually be extracted coincide only after the allowed operations and the available reference resources are fixed [1–3]. Quantum batteries make this distinction concrete: one cares not only how much energy is stored, but how fast a useful nonpassive state can be prepared [4, 5]. Repeated interactions give a natural continuous setting, in which a system collides with a stream of fresh probes [6–9].

Long-lived coherence in a V-type three-level system has recently been studied as a work resource [10]. Our question is sharper. *If that coherence is renewed between collisions, can it charge a battery tape continuously, and what limits the power?* In particular: is there a simple matching condition between the rate at which the bath rebuilds coherence and the rate at which collisions remove it?

We answer this with a minimal model. A V-type system with a coherent excited doublet interacts with ground-state ancillas through a matched-gap exchange. The collision itself needs no external work. We derive the exact channel and the fixed point, find the power optimum, and quantify the penalties from finite precession and finite coupling. We also separate total ergotropy from work that is accessible under stated discharge rules. Long-lived V-system coherence is the resource; the continuous, stroboscopic conversion law and its impedance-matched power optimum are our contribution.

We report total ergotropy as an upper bound on battery work capacity. Without a phase reference, the coherent part of a single outgoing ancilla is not freely extractable by energy-preserving operations alone; that distinction is made precise in Sec. V.

II. MATCHED-GAP COLLISION MODEL

The system basis is $\{|0\rangle, |1\rangle, |2\rangle\}$. We use the doublet variables

$$s = \rho_{11} + \rho_{22}, \qquad d = \rho_{11} - \rho_{22}, \qquad C = \rho_{12}. \quad (1)$$

The free Hamiltonians are matched at a finite gap Δ,

$$H_S = \bar{E}(|1\rangle\langle 1| + |2\rangle\langle 2|) + \frac{\Delta}{2}(|1\rangle\langle 1| - |2\rangle\langle 2|), \quad (2)$$

$$H_A = \Delta|\uparrow\rangle\langle\uparrow|. \quad (3)$$

* sai@phy.iitb.ac.in
† bm@ee.iitb.ac.in

A fresh ancilla has excited population α and no coherence. During a collision of duration τ,

$$H_{\rm int} = \hbar g\big(|1\downarrow\rangle\langle 2\uparrow| + |2\uparrow\rangle\langle 1\downarrow|\big), \qquad \theta = g\tau. \tag{4}$$

On the active block, $H_S + H_A$ is proportional to the identity, so $[H_S+H_A, H_{\rm int}] = 0$. The same exchange conserves the intended excitation number. A finite matched $\Delta > 0$ is essential. At exact doublet degeneracy, strict resonance would force a gapless ancilla, which cannot store energetic ergotropy.

Tracing the exact unitary $U_\theta = \exp(-iH_{\rm int}\tau/\hbar)$ yields

$$s' = s, \tag{5a}$$
$$d' = \cos^2\theta\, d - \sin^2\theta(1-2\alpha)s, \tag{5b}$$
$$C' = \cos\theta\, C. \tag{5c}$$

The outgoing ancilla Bloch components are

$$p'_\uparrow = \alpha + \frac{\sin^2\theta}{2}\big[s(1-2\alpha) + d\big], \tag{6a}$$
$$\rho'_{\uparrow\downarrow} = -i\sin\theta(1-2\alpha)C, \tag{6b}$$
$$z_A = 2p'_\uparrow - 1, \quad x_A = 2\Re\rho'_{\uparrow\downarrow}, \quad y_A = -2\Im\rho'_{\uparrow\downarrow}. \tag{6c}$$

Direct 6×6 unitary propagation agrees with Eqs. (5)–(6) to floating-point precision (Fig. 5).

III. RENEWAL AND STROBOSCOPIC COHERENCE

We use two renewal pictures. The analytic benchmark is an affine, phase-locked reset over the waiting time $T = 1/r$:

$$C_{n+1} = q_C e^{-i\phi}\cos\theta\, C_n + (1-q_C)c_0,$$
$$q_C = e^{-\gamma_C/r}, \qquad \phi = \Delta/r. \tag{7}$$

Its fixed point is exact:

$$C^* = \frac{(1-q_C)c_0}{1 - q_C e^{-i\phi}\cos\theta}. \tag{8}$$

Population relaxation is treated by a separate factor $q_d = e^{-\gamma_d/r}$ and is kept in the numerical curves.

A lab-frame steady coherence at finite splitting is not automatic for a single undriven thermal bath. The affine map must be justified. We do this with a directed dark-state-leakage bath: bright-state pumping and relaxation $(\Gamma_\uparrow, \Gamma_\downarrow)$ plus dark leakage at rate κ_D,

$$\dot p_B = \Gamma_\uparrow p_0 - \Gamma_\downarrow p_B, \qquad \dot p_D = -\kappa_D p_D, \qquad p_0 = 1 - p_B - p_D. \tag{9}$$

Its steady state is $p_D^{\rm ss} = 0$ and $p_B^{\rm ss} = \Gamma_\uparrow/(\Gamma_\uparrow + \Gamma_\downarrow) \equiv s_{\rm ss}$. The real doublet coherence is the bright–dark imbalance, $C = (p_B - p_D)/2$, so the bath stabilizes

$$c_0 = \frac{s_{\rm ss}}{2} \tag{10}$$

by itself. When the rates match, $\kappa_D = \Gamma_\uparrow + \Gamma_\downarrow \equiv \gamma_C$, the waiting-time map of Eq. (9) reduces to the affine reset with $q_C = e^{-\gamma_C/r}$ and $c_0 = s_{\rm ss}/2$, up to $O(\theta^2)$ corrections from the dark population made by the collision (Appendix B). The power optimum sits in this weak-collision regime, so the main formulas [Eqs. (8) and (15)] use microscopically fixed γ_C and c_0. Away from matched rates we keep the full two-dimensional affine map and solve it numerically (Fig. 4). The phase $e^{-i\phi}$ is free precession at the finite gap. The fully autonomous, phase-unreferenced reading of our results is safest for $\delta = \Delta/\gamma_C \lesssim 1$, where Eq. (15) already gives the precession penalty. Symmetric bright–dark mixing is *not* a renewal mechanism: it equalizes the two populations and kills C.

IV. ERGOTROPY AND IMPEDANCE MATCHING

For a qubit with Hamiltonian $H_A = \Delta|\uparrow\rangle\langle\uparrow|$, Bloch length $|\boldsymbol{r}_A|$, and vertical component z_A, the ergotropy is

$$\mathcal{W}_A = \frac{\Delta}{2}\big(z_A + |\boldsymbol{r}_A|\big). \tag{11}$$

In the weak-extraction, ground-biased regime,

$$\mathcal{W}_A \simeq \frac{\Delta\sin^2\theta\,|C^*|^2}{|z_A|}, \qquad P_{\rm erg} = r\mathcal{W}_A. \tag{12}$$

Define the extraction rate

$$\Gamma_{\rm ext} = r(1-\cos\theta). \tag{13}$$

At small precession, maximizing Eq. (12) gives a simple answer:

$$\Gamma_{\rm ext} \simeq \gamma_C. \tag{14}$$

Extraction faster than renewal depletes C^*. Extraction slower than renewal wastes regenerated coherence. That is the impedance match. For $\delta = \Delta/\gamma_C$, the small-angle optimum and bound become

$$r_{\rm opt} \simeq \frac{\gamma_C\sqrt{1+\delta^2}}{1-\cos\theta}, \tag{15a}$$
$$P_{\rm erg}^{\rm max}(\delta) \simeq \frac{\gamma_C\Delta c_0^2}{|z_A|}\frac{1}{1+\sqrt{1+\delta^2}}. \tag{15b}$$

So $P_{\max}(0) = \gamma_C\Delta c_0^2/(2|z_A|)$, and the normalized finite-phase suppression is $2/[1+\sqrt{1+\delta^2}]$.

At zero phase the finite-angle harvesting prefactor is almost flat,

$$\mathcal{C}(\theta) = \frac{1}{2} - \frac{\theta^4}{96} + O(\theta^6). \tag{16}$$

The formal limit $\theta \to 0$, $r \to \infty$ is not free: $\theta = g\tau \le g/r$. At the optimum this requires

$$\theta \gtrsim \frac{2\gamma_C\sqrt{1+\delta^2}}{g}. \tag{17}$$

The bound is linear, not square-root, in γ_C/g.

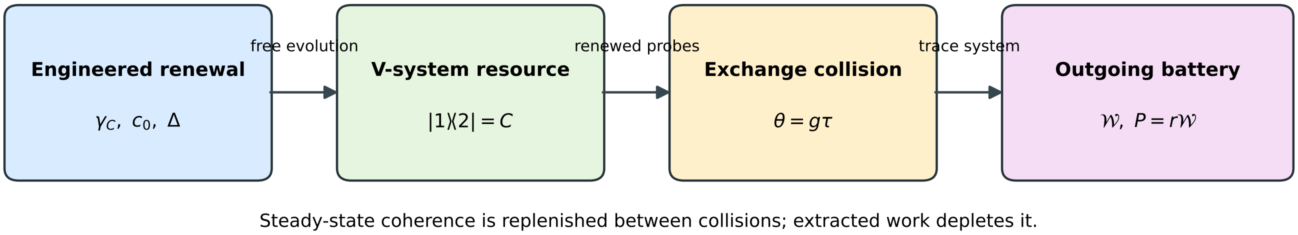

FIG. 1. Renewal rebuilds the V-system doublet coherence between exchange collisions. Independent outgoing qubits carry energy and ergotropy. The analytic reset is phase locked. The directed-leakage bath gives an autonomous comparison in the weak-collision regime.

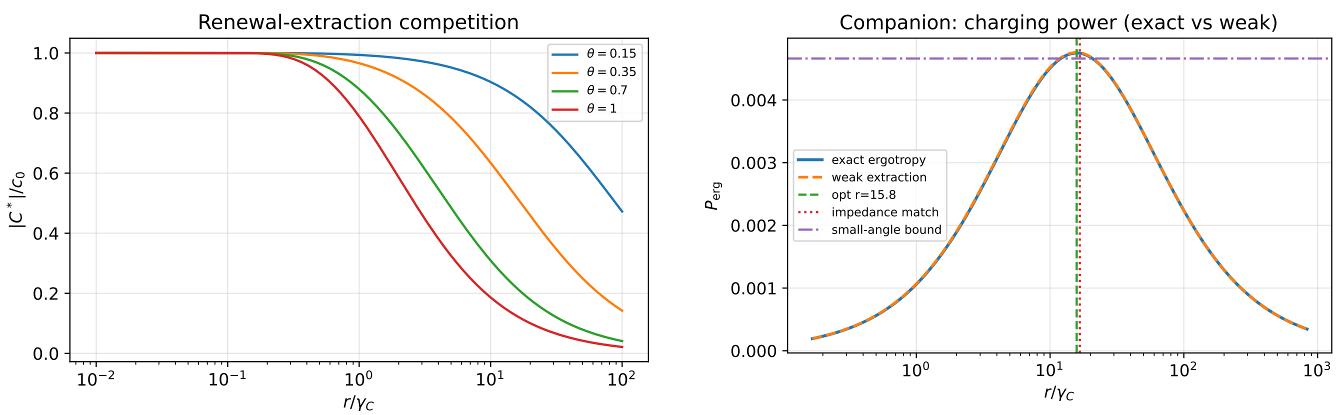

FIG. 2. Left: fixed-point coherence under renewal and collisional depletion at zero precession. Right: exact power and weak-extraction approximation for $(\gamma_C, \gamma_d, c_0, s_{\rm ss}, \Delta, \theta, \alpha) = (1, 1, 0.25, 0.5, 0.15, 0.35, 0)$.

V. ACCESSIBLE WORK

Equation (11) is the maximum under unrestricted unitaries. Without a phase reference, time-translation-covariant one-copy discharge first dephases between distinct energies. What remains is

$$\mathcal{W}_{\rm acc}^{(1)} = \mathcal{W}_{\rm inc}. \tag{18}$$

For the purely coherent, ground-biased outputs we study, this is zero. An explicit phase reference can unlock the total ergotropy, but then the reference itself is part of the resource budget [3].

Two copies keep relational coherence inside the one-excitation sector. Write one output as

$$\rho_A = \begin{pmatrix} a & c \\ c^* & b \end{pmatrix}, \qquad a + b = 1, \quad a \geq b. \tag{19}$$

After total-energy dephasing of $\rho_A^{\otimes 2}$, eigenvalue sorting gives

$$\mathcal{W}_{\rm acc}^{(2)} = \begin{cases} 0, & |c|^2 \leq b(a-b), \\ \Delta[|c|^2 - b(a-b)], & b(a-b) < |c|^2 \leq a(a-b), \\ \Delta[2|c|^2 - (a-b)], & a(a-b) < |c|^2 \leq ab. \end{cases} \tag{20}$$

The weak-harvesting curves usually stay below the first threshold. We therefore report total ergotropy as an upper bound, not as work that is automatically available from one or two batteries.

VI. DISCUSSION

The model shows a simple rate-matching rule: the gross charging power is largest when collisional depletion and coherence regrowth are comparable. Finite-angle numerics keep this picture. Precession only multiplies the bound by a closed dimensionless factor. In the weak-collision window where the optimum sits, the affine map, γ_C, and c_0 all follow from the directed dark-state-leakage bath (Appendix B). Away from matched rates the power degrades, as shown by the leakage scan, and physicality

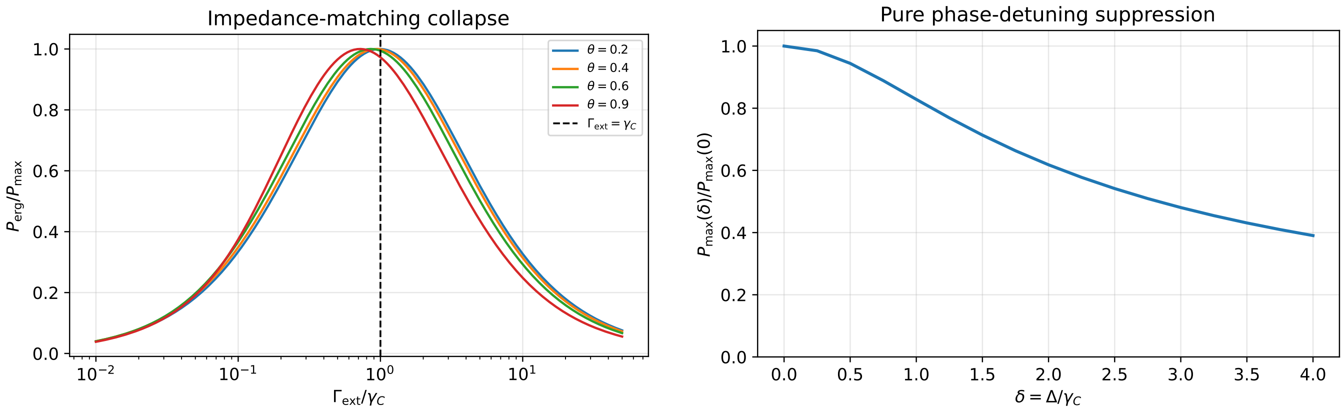

FIG. 3. Left: normalized exact power collapses around $\Gamma_{\rm ext}/\gamma_C = 1$. Right: finite-precession suppression with the energy prefactor held fixed, so the plot isolates phase detuning from the linear factor of Δ in the ergotropy.

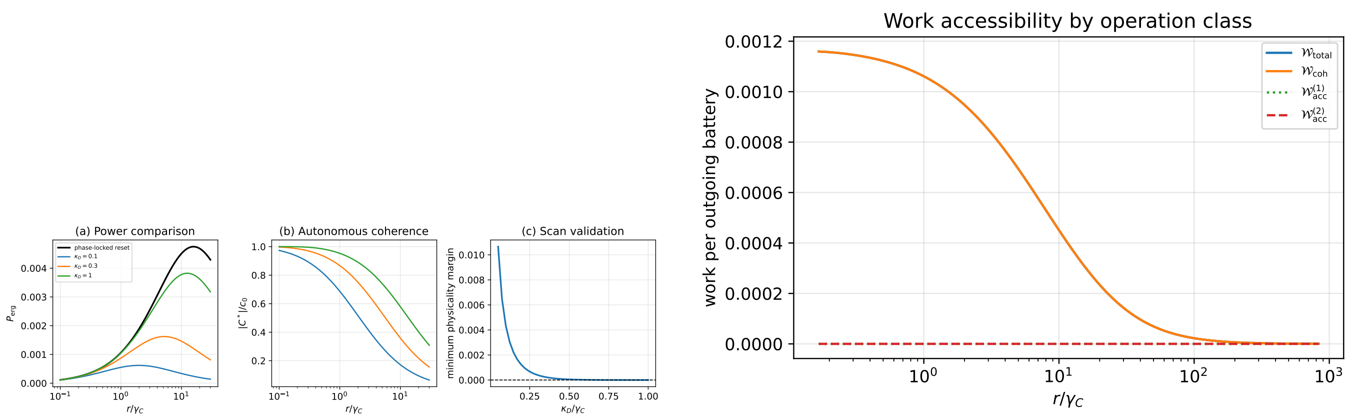

FIG. 4. Left: phase-locked reset versus directed dark leakage, including the retained coherence and the minimum physicality margin over the scanned rates. Right: total and coherent ergotropy compared with one- and two-copy work after the stated energy dephasings.

checks keep unphysical coherence out of the comparison.

Some costs sit outside the gross power in Eq. (15). Ground-state ancillas are low-entropy resources; their preparation free energy must enter any net efficiency. A phase reference used at discharge is likewise a resource. These points do not change the collision map or the impedance match. They do limit the thermodynamic claim.

A transport-driven spin valve can regenerate coherence and also produce electrical power. That extension needs a lead-resolved nonsecular generator, positivity checks, and a real electrical-power/ergotropy tradeoff. It is not part of the present companion result.

VII. CONCLUSION

A resonant, charge-preserving exchange converts renewed V-system coherence into a stream of nonpassive qubit states. The stroboscopic solution gives impedance matching, a flat finite-angle plateau, and a finite-precession penalty. The operational statement is narrower than the gross ergotropy alone: without a phase reference the one-copy coherent part is locked, and two-copy activation follows the thresholds in Eq. (20). These distinctions put the coherence-harvesting bound on a clear thermodynamic footing.

ACKNOWLEDGMENTS

The idea of coupling an ancilla tape to a coherence-generating open system, in order to harvest coherence as battery ergotropy, was conceived by S.V. and B.M. N.J. developed the companion model and the analytic and numerical results. All authors contributed to checking and improving the results and the manuscript.

Appendix A: Unitary channel check

In the ordered active basis $\{|1\downarrow\rangle, |2\uparrow\rangle\}$,

$$U_\theta = \begin{pmatrix} \cos\theta & -i\sin\theta \\ -i\sin\theta & \cos\theta \end{pmatrix}. \tag{A1}$$

Embedding this block in the six-dimensional product space and tracing the ancilla reproduces Eq. (5). The validation samples physical doublet states and random θ, α; the error data are exported with the figures.

Appendix B: Reduction of directed leakage to the affine renewal

With $A = \Gamma_\uparrow + \Gamma_\downarrow$, $q_B = e^{-A\tau_c}$, $q_D = e^{-\kappa_D \tau_c}$, the waiting-time map of Eq. (9) from post-collision values (B_+, D_+) is

$$D_- = q_D D_+, \qquad B_- = q_B B_+ + s_{\rm ss}(1-q_B) - \Gamma_\uparrow D_+ \frac{q_D - q_B}{A - \kappa_D}, \tag{B1}$$

while the collision acts as $(B_+, D_+)^T = R_\theta (B_-, D_-)^T$ with $R_\theta = \frac{1}{2}\left(\begin{smallmatrix} 1+\cos\theta & 1-\cos\theta \\ 1-\cos\theta & 1+\cos\theta \end{smallmatrix}\right)$. At matched rates $\kappa_D = A \equiv \gamma_C$ one has $q_B = q_D = q_C$ and the cross term becomes $\Gamma_\uparrow D_+ \tau_c e^{-\gamma_C \tau_c}$. The collision generates dark population only at second order, $D_+ = \frac{1-\cos\theta}{2}(B_- - D_-) = O(\theta^2)$, so to leading order in θ the imbalance $C = (B-D)/2$ obeys

$$C_{n+1} = q_C \cos\theta \, C_n + (1 - q_C)\,\frac{s_{\rm ss}}{2}, \tag{B2}$$

which is the affine benchmark with $\gamma_C = \kappa_D = A$ and $c_0 = s_{\rm ss}/2$. Because the power optimum obeys $1-\cos\theta \simeq \gamma_C/r \ll 1$, the neglected terms enter the fixed point at relative order θ^2 and the optimal power at order θ^4, in line with the flat plateau. The mismatched-rate case is the two-dimensional affine map used for Fig. 4.

Appendix C: Two-copy eigenvalue ordering

Total-energy dephasing leaves populations a^2 and b^2 at energies 0 and 2Δ, and a one-excitation block

$$\begin{pmatrix} ab & |c|^2 \\ |c|^2 & ab \end{pmatrix} \tag{C1}$$

with eigenvalues $ab \pm |c|^2$. Pairing decreasing eigenvalues with the energy list $\{0, \Delta, \Delta, 2\Delta\}$ yields Eq. (20). The boundaries are $ab - |c|^2 = b^2$ and $ab + |c|^2 = a^2$. Positivity enforces $|c|^2 \le ab$.

[1] A. E. Allahverdyan, R. Balian, and T. M. Nieuwenhuizen, Maximal work extraction from finite quantum systems, Europhysics Letters **67**, 565 (2004).

[2] G. Francica, F. C. Binder, G. Guarnieri, M. T. Mitchison, J. Goold, and F. Plastina, Quantum coherence and ergotropy, Physical Review Letters **125**, 180603 (2020).

[3] M. Lostaglio, D. Jennings, and T. Rudolph, Description of quantum coherence in thermodynamic processes requires constraints beyond free energy, Physical Review X **5**, 021001 (2015).

[4] R. Alicki and M. Fannes, Entanglement boost for extractable work from ensembles of quantum batteries, Physical Review E **87**, 042123 (2013).

[5] F. Campaioli, F. A. Pollock, F. C. Binder, L. C. Céleri, J. Goold, S. Vinjanampathy, and K. Modi, Enhancing the charging power of quantum batteries, Physical Review Letters **118**, 150601 (2017).

[6] L.-P. Bettmann, J. Goold, and M. T. Mitchison, Repeated-interaction description of a monitored double-quantum-dot heat engine, Physical Review E **107**, 044102 (2023), arXiv:2212.10193.

[7] A. Prositto, B. Forbes, and D. Segal, Equilibrium and nonequilibrium steady states with the repeated interaction protocol: relaxation dynamics and energetic cost (2025), arXiv:2501.05392.

[8] N. Massa, F. Cavaliere, and D. Ferraro, Collisional charging of a transmon quantum battery, Batteries **11**, 240 (2025), arXiv:2506.16177.

[9] G. Barsky-Giles, A. Prositto, M. Gerry, and D. Segal, Impossibility of refrigeration and engine operation in minimal qubit repeated-interaction models (2026), arXiv:2602.18300, arXiv:2602.18300.

[10] W. Chen, S.-W. Han, X. Feng, and J. Feng, Work extraction from long-lived quantum coherence of a three-level system (2025), arXiv:2508.02614.

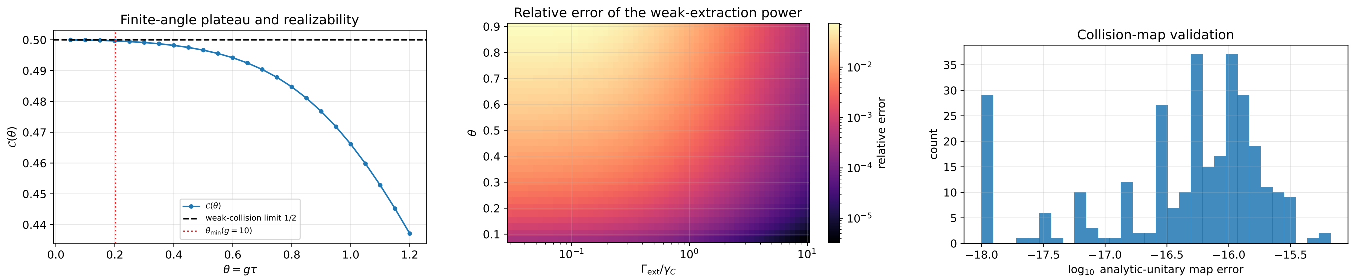

FIG. 5. Checks beyond the main power curves. Left: finite-angle plateau and realizability threshold. Center: accuracy of the weak-extraction formula. Right: analytic-channel error against direct unitary propagation for random physical input states.